

AI-SA-IN-TH-000016

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Understanding Hallucinations

Types, Root Causes, Risk Factors, and Why They Cannot Be Eliminated

9 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

Theory

Domain

Learning Objectives

- 1 Define hallucination precisely and distinguish it from factual errors made by humans
- 2 Explain why token prediction architecture makes hallucination an inherent, not incidental, property of LLMs
- 3 Classify a given LLM failure into one of five hallucination types
- 4 Describe the overconfidence problem and explain why it makes hallucinations especially dangerous
- 5 Assess the hallucination risk of a deployment scenario using a structured domain risk framework
- 6 Identify ten signals in a prompt or system design that increase hallucination probability

What Is a Hallucination?

LLM Hallucination

A hallucination is a model output that is grammatically fluent and contextually plausible but factually incorrect, fabricated, or unsupported by any real-world evidence — generated with apparent confidence by a language model that has no mechanism to verify truth.

- Coined by analogy with human perceptual hallucination: the model perceives (outputs) something that isn't there
- Key property: the model does not 'know' the output is wrong — there is no internal fact-checking mechanism
- Distinct from lying: the model has no intent to deceive; it produces the statistically most likely continuation
- Distinct from a mistake: a human mistake involves incorrect reasoning over correct facts; a hallucination involves generating facts that never existed
- Spectrum: ranges from subtly wrong (wrong date, swapped name) to completely fabricated (non-existent court case, invented study)

Root Cause — Token Prediction Architecture

Training Corpus

Internet text: contains facts, fiction, errors, opinions
— model cannot distinguish

Next-Token Prediction

Maximise $P(\text{token}_t | \text{context})$
No truth-checking objective
Fluency $\neq$ accuracy

Statistical Pattern

Model learns co-occurrence statistics, not causal world knowledge

Hallucination Output

Plausible-sounding but factually wrong; stated with confidence

Fundamental insight: the training objective (predict next token) is misaligned with the deployment goal (produce accurate facts). This gap cannot be closed by more data or bigger models alone.

Why elimination is impossible: to eliminate hallucinations entirely, the model would need a real-time oracle of world facts — which doesn't exist. RAG and grounding reduce risk but cannot reach zero.

Five Types of Hallucination

Type	Definition	Example	Risk Level
Intrinsic	Output contradicts or distorts information explicitly present in the source/context	Summarising a document but attributing wrong numbers or wrong person's quote	High — directly verifiable against input
Extrinsic	Output introduces information not present in the source/context and cannot be verified	Citing a statistic not in the retrieved document — fabricated from model's 'memory'	Very High — invisible without external verification
Temporal	Output states facts that were true at training time but are now outdated	Stating a now-replaced CEO, repealed law, or superseded scientific finding	Medium — time-indexed verification needed
Entity	Swapping, confusing, or fabricating named entities (people, places, organisations)	Attributing a quote to the wrong author; wrong birth year for a historical figure	High — names and dates are hard to verify
Factual	Generating entirely false claims about the world with no basis in training data	Inventing a court case, fabricating a published study, citing non-existent statistics	Critical — most dangerous in high-stakes domains

Hallucination Examples — Legal, Medical, Scientific

Three high-profile real-world cases demonstrating hallucination in consequential domains

Legal — Mata v. Avianca (2023)

Attorney Steven Schwartz used ChatGPT to prepare a legal brief. ChatGPT generated six plausible legal case citations — complete with docket numbers, judges' names, and quotations — none of which existed. When the judge asked for the actual cases, Schwartz could not produce them. He was sanctioned \$5,000. ChatGPT had stated 'Yes, these are real cases' when asked to verify them.

Type: Factual + Entity hallucination. Mechanism: model pattern-matched legal citation format from training data without any grounding in an actual legal database.

Medical / Scientific — Drug Dosage and Citation Errors

Studies (Shah et al., 2023; Thirunavukarasu et al., Lancet Digital Health 2023) found that GPT-4 fabricated medical references in 68.4% of generated citations when asked for medical literature, and produced incorrect drug dosage information in ~26% of queries tested without RAG grounding.

Type: Extrinsic (fabricated citations) + Factual (wrong dosages). Risk: directly dangerous if acted upon without verification by a licensed clinician.

Closed-Domain vs Open-Domain Hallucination

Closed-Domain

- Model operates on a provided document, context, or database
- Hallucination occurs when model contradicts or extends beyond the given source
- Also called 'faithfulness hallucination'
- Detectable by comparing output to input source
- Lower risk: grounding documents provide a verification baseline
- Common in: summarisation, document QA, RAG systems
- Example: summarising a financial report but inventing a revenue figure not in the document

Open-Domain

- Model answers freely from parametric knowledge (memorised during training)
- No context document to compare against
- Also called 'factual hallucination'
- Only detectable via external knowledge bases or human expert review
- Higher risk: no automatic verification baseline exists
- Common in: general Q&A, conversational AI, knowledge retrieval
- Example: asked to describe a historical event, model invents plausible-sounding but false details

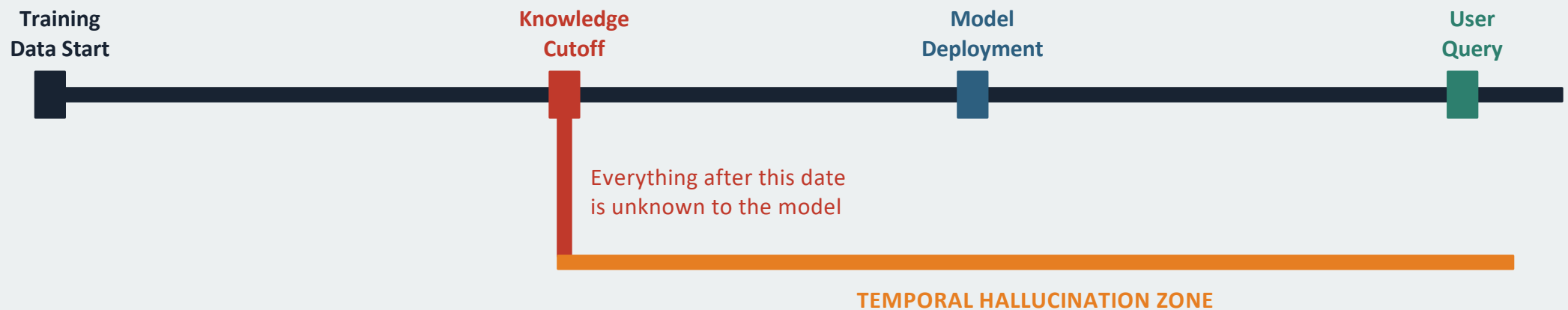
The Overconfidence Problem

Hallucination Overconfidence

LLMs systematically state hallucinated information with the same linguistic confidence as accurate information — without hedging, uncertainty markers, or reduced fluency. This makes hallucinations uniquely dangerous: a wrong answer sounds identical to a correct one.

- Calibration failure: a well-calibrated model would say 'I'm not sure' for low-knowledge claims; LLMs are miscalibrated — expressed confidence does not predict factual accuracy
- Fluency as false signal: users perceive fluent, well-structured text as reliable; hallucinations are often more fluent than hedged correct answers
- Authoritative framing: models trained on expert text learn to write in an authoritative register — 'Research shows that...' 'According to...', even when fabricating
- Expected Calibration Error (ECE): measures gap between stated confidence and actual accuracy; LLMs have $ECE > 0.15$ on factual QA tasks without calibration
- Mitigation: require uncertainty expressions ('I am not certain...'), use temperature=0 for factual tasks, implement confidence scoring as a guardrail layer

Knowledge Cutoff — How Temporal Hallucinations Arise



Model will answer questions about this period using outdated knowledge — stating replaced CEOs, superseded policies, recalled products as current facts.

Mitigation: always display knowledge cutoff date to users; use web search tool for current events; add 'as of [date]' to responses involving time-sensitive facts.

Hallucination Rates by Domain and Model Tier

Domain	Unmitigated Hallucination Rate	With RAG Grounding	Risk Tier	Primary Type
Medical / Clinical	26–40%	< 5% (with citation enforcement)	Critical	Factual + Extrinsic
Legal	35–45%	< 8% (with legal DB RAG)	Critical	Entity + Factual
Financial	20–35%	< 6% (with structured data RAG)	High	Factual + Temporal
Scientific Research	40–68%	< 10% (with PubMed RAG)	High	Extrinsic (citation)
General Knowledge	15–25%	< 8% (with web search tool)	Medium	Temporal + Factual
Code Generation	10–20%	N/A (tested by execution)	Medium	Intrinsic (wrong API)

Why Hallucinations Cannot Be Completely Eliminated

The Architectural Impossibility

Hallucinations are an inherent property of probabilistic language models, not a fixable bug. They arise from the fundamental mismatch between the training objective (predict likely text) and the deployment goal (produce accurate information). Three structural reasons explain why zero-hallucination is impossible.

- Reason 1 — No truth oracle: the model has no mechanism to check whether a generated token corresponds to a real-world fact; it only knows statistical co-occurrence patterns
- Reason 2 — Finite training data: any training corpus is an incomplete sample of world knowledge; knowledge gaps cause the model to generalise (hallucinate) beyond what it has seen
- Reason 3 — Probability distribution tails: even a well-calibrated model assigns non-zero probability to incorrect tokens; at scale, rare hallucinations become frequent events
- What we can do: reduce hallucination rate to acceptable levels through grounding (RAG), detection (SelfCheckGPT, NLI), and human review for high-stakes outputs
- Target: 'acceptable' hallucination rate is domain-dependent — < 1% for medical, < 5% for legal, < 15% for general consumer; define your target before deployment

Case Study — Medical Chatbot Hallucination Harm

Case: AI-Powered Medical Information Chatbot — Production Deployment (UK, 2024)

- A UK private healthcare provider deployed an LLM chatbot for patient triage and medication information — accessible 24/7 without mandatory physician review
- A patient asked about maximum safe dose of ibuprofen for a post-surgical patient taking blood thinners (warfarin) — a known dangerous combination
- The chatbot stated: 'For post-surgical pain, ibuprofen 400mg up to three times daily is generally safe' — omitting the critical warfarin interaction entirely
- The patient followed this advice; was hospitalised two days later with internal bleeding from the ibuprofen-warfarin interaction
- Post-incident analysis: the model had no access to patient medication records; was not grounded in real-time drug interaction databases; had no mandatory physician escalation for high-risk drug queries

Key Lesson

Prevention required three changes: (1) RAG grounding against BNF (British National Formulary) drug interaction database; (2) mandatory query classification to detect drug-drug interaction questions; (3) hard rule: all drug combination queries must escalate to human pharmacist regardless of model confidence. This incident triggered UK MHRA guidance on medical AI chatbot requirements.

Factual vs Faithfulness Hallucination

Factual Hallucination

- Occurs in open-domain settings: model generates false information from parametric memory
- Cannot be verified against any input document — requires external knowledge base
- Example: 'The Eiffel Tower was built in 1887' (actual: 1889) — no document in context to check against
- Harder to detect: requires external fact-checking, knowledge graphs, or human expert review
- Primary mitigation: tool-use (Wolfram Alpha, web search), knowledge base grounding, temporal citations
- Benchmark: TruthfulQA measures this type

Faithfulness Hallucination

- Occurs in closed-domain settings: model contradicts or adds to the provided context document
- Can be verified against the input document — automated detection is feasible
- Example: summarising a report but attributing revenue growth to the wrong quarter
- Easier to detect: NLI models can check if the output is entailed by the input context
- Primary mitigation: citation enforcement, faithfulness scoring (RAGAS), NLI consistency check
- Benchmark: RAGAS faithfulness metric, SummaC measure this type

Domain Risk Score — Worked Assessment

$$\text{Hallucination Risk Score} = \text{Harm Severity} \times \text{Hallucination Rate} \times (1 - \text{Mitigation Coverage})$$

1. Scale: Harm Severity (1–5): 1=cosmetic, 3=financial loss, 5=physical harm or death
2. Hallucination Rate: unmitigated domain baseline (see Table, Slide 10)
3. Mitigation Coverage: fraction of hallucination risk covered by implemented controls (0–1)
4. Example — Medical chatbot without RAG: Score = $5 \times 0.33 \times (1 - 0.0) = 1.65 \rightarrow \text{CRITICAL } (> 1.0)$
5. Example — Medical chatbot with RAG + citation enforcement: Score = $5 \times 0.33 \times (1 - 0.85) = 0.25 \rightarrow \text{ACCEPTABLE } (< 0.3)$
6. Example — General consumer chatbot with web search: Score = $2 \times 0.20 \times (1 - 0.60) = 0.16 \rightarrow \text{ACCEPTABLE}$

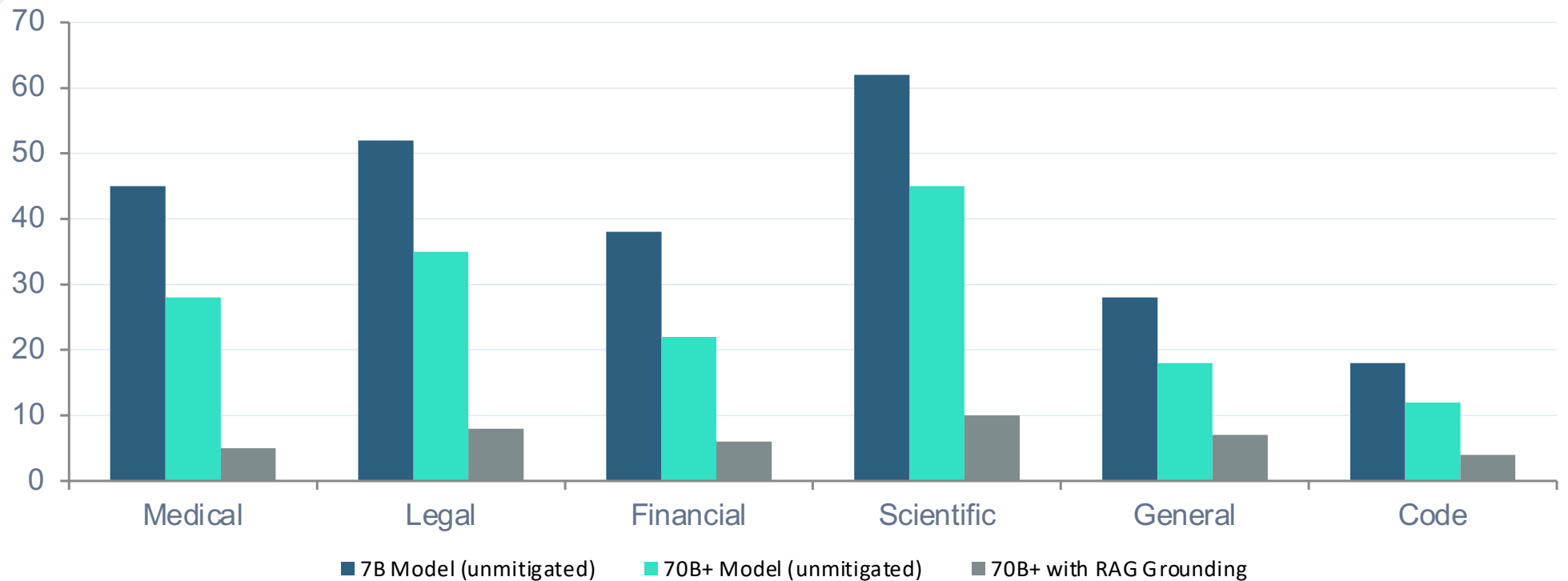
Note: Deployment gate: Risk Score $> 1.0 \rightarrow$ deployment blocked pending additional mitigation. $0.3\text{--}1.0 \rightarrow$ deployment with HITL. $< 0.3 \rightarrow$ automated deployment permitted with monitoring.

Ten Signals That Increase Hallucination Risk

Design or prompt characteristics that correlate with elevated hallucination probability:

- Long-tail knowledge queries: asking about obscure facts, rare diseases, or niche topics the model has seen few training examples of
- Numerical precision requests: asking for exact statistics, percentages, dates, or measurements — numbers hallucinate more than prose
- Citation or source requests: 'Provide three references for...' — models fabricate plausible-format citations when sources aren't in context
- Temporal specificity: queries anchored to recent dates or events near or after the training cutoff
- Complex multi-hop reasoning: tasks requiring the model to chain many facts — each hop multiplies hallucination probability
- Contradictory or ambiguous context: when retrieved documents conflict, models often hallucinate to 'resolve' the contradiction
- Very long context windows: attention quality degrades over long contexts — facts from the middle of a long context are more likely to be misremembered
- High-specificity instructions: 'Give exact legal citations for...' forces the model to commit to specific claims it cannot verify
- Creative framing for factual queries: wrapping factual questions in story/roleplay contexts lowers the model's guard against factual errors
- No grounding documents provided: open-domain factual queries without RAG context are the highest-risk configuration

Hallucination Rate by Domain and Model Size



Source: aggregated from multiple evaluation studies (Shah et al. 2023, Thirunavukarasu et al. 2023, Ji et al. 2023). Larger models reduce but do not eliminate hallucination. RAG grounding provides 5–7× reduction across all domains.

Key Takeaways

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- 1 Hallucination is architectural, not a bug: LLMs predict statistically likely tokens, not true tokens — this fundamental mismatch between training objective and deployment goal means zero hallucination is impossible
- 2 Five types require different mitigations: intrinsic/faithfulness hallucinations (model contradicts source) are detectable via NLI; factual/extrinsic hallucinations (model fabricates) require external grounding and fact-checking
- 3 The overconfidence problem makes hallucinations uniquely dangerous: hallucinated content sounds as fluent and confident as accurate content — users cannot detect it without independent verification
- 4 Hallucination rates vary dramatically by domain: unmitigated rates range from 12% (code) to 68% (scientific citations); RAG grounding reduces rates by 5–7× but does not reach zero
- 5 Domain risk assessment must precede deployment: compute Risk Score = Harm Severity × Rate × (1 – Mitigation); scores above 1.0 require additional mitigation before production launch